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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 20th July Evening Shift

JEE Main Question Paper with Answer Key

20 Jul 2021	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	24 questions
Mathematics	21 questions
Physics	27 questions
Out of syllabus	18 questions
Answer key	at the end

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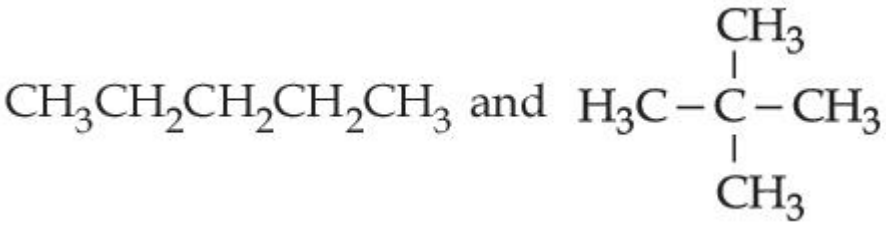
Chemistry 24 questions

Q1. MCQ (Single Correct Answer)

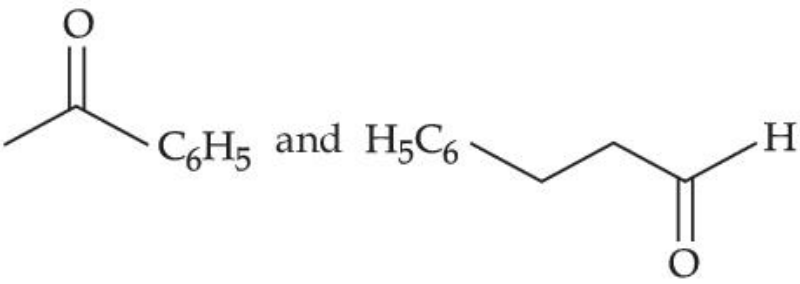
Open solution

Which one of the following pairs of isomers is an example of metamerism?

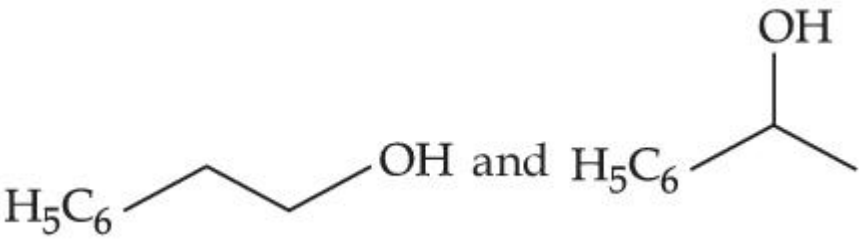
(A)



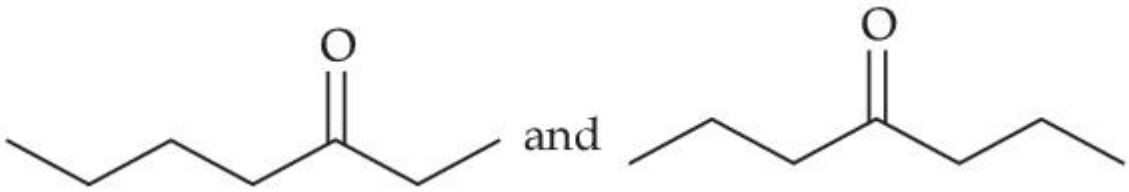
(B)



(C)



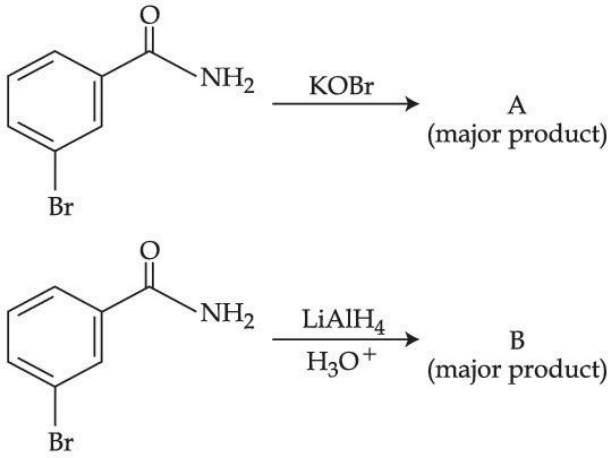
(D)



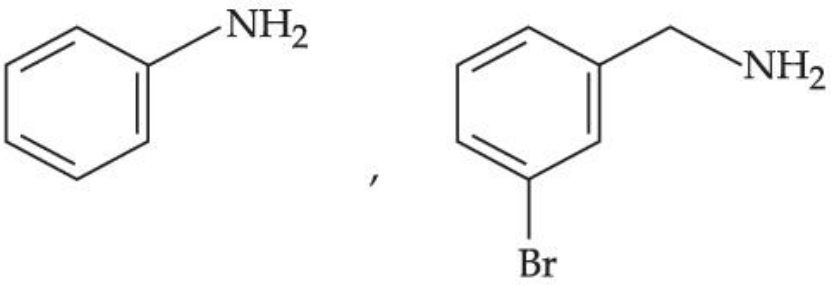
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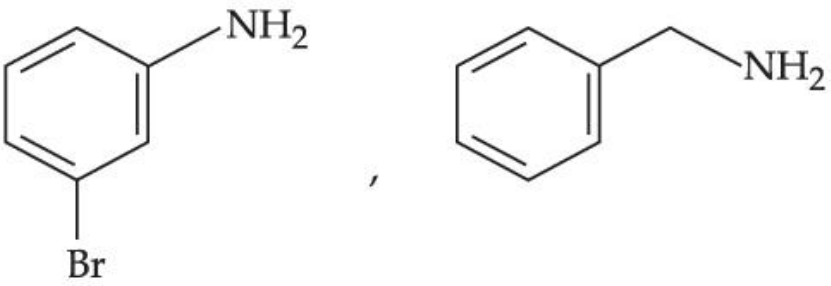
Q2. MCQ (Single Correct Answer)

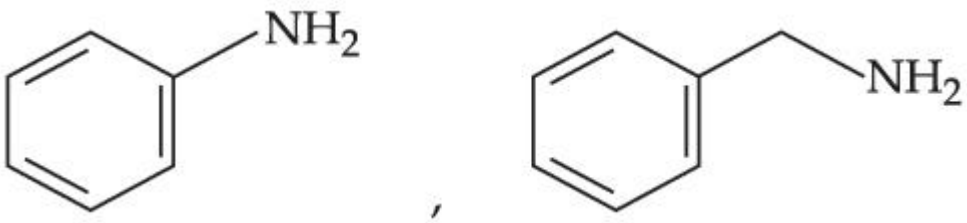
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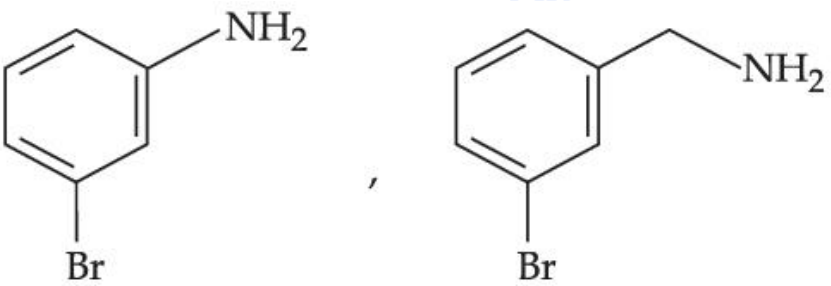


In the above reactions, product A and product B respectively are :

(A) 

(B) 

(C) 

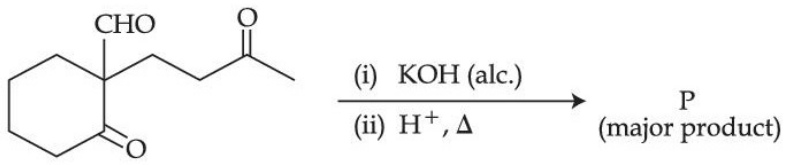
(D) 

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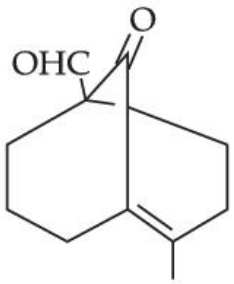
Q3. MCQ (Single Correct Answer)

Open solution

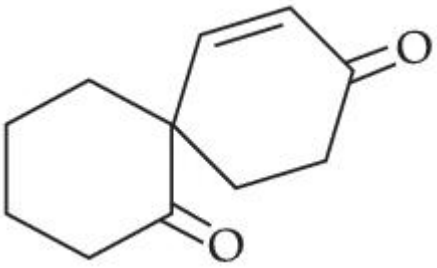
The major product (P) in the following reaction is :



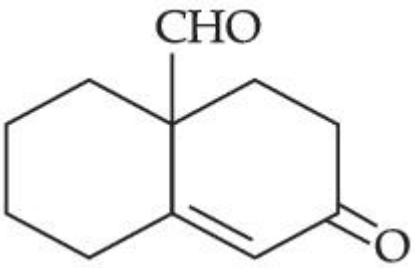
(A)



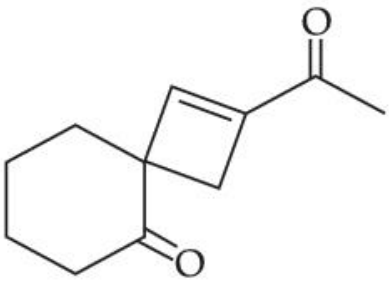
(B)



(C)



(D)



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Q4. MCQ (Single Correct Answer)

A solution is 0.1 M in Cl^- and 0.001 M in CrO_4^{2-}. Solid $AgNO_3$ is gradually added to it. Assuming that the addition does not change in volume and $K_{sp}(AgCl) = 1.7 \times 10^{-10}\ M^2$ and $K_{sp}(Ag_2CrO_4) = 1.9 \times 10^{-12}\ M^3$.

Select correct statement from the following :

(A) AgCl precipitates first because its K_{sp} is high.

(B) Ag_2CrO_4 precipitates first as its K_{sp} is low.

(C) Ag_2CrO_4 precipitates first because the amount of Ag^+ needed is low.

(D) AgCl will precipitate first as the amount of Ag^+ needed to precipitate is low.

Open solution

Q5. MCQ (Single Correct Answer)

Outermost electronic configuration of a group 13 element, E, is $4s^2, 4p^1$. The electronic configuration of an element of p-block period-five placed diagonally to element, E is :

(A) $[Kr]\ 3d^{10}\ 4s^2\ 4p^2$

(B) $[Ar]\ 3d^{10}\ 4s^2\ 4p^2$

(C) $[Xe]\ 5d^{10}\ 6s^2\ 6p^2$

(D) $[Kr]\ 4d^{10}\ 5s^2\ 5p^2$

Open solution

Q6. MCQ (Single Correct Answer)

Metallic sodium does not react normally with :

(A) gaseous ammonia

(B) But-2-yne

(C) Ethyne

(D) tert-butyl alcohol

Open solution

Q7. MCQ (Single Correct Answer)

Spin only magnetic moment of an octahedral complex of Fe^{2+} in the presence of a strong field ligand in BM is :

(A) 4.89

(B) 2.82

(C) 0

(D) 3.46

Open solution

Q8. MCQ (Single Correct Answer)

Which one of the following species doesn't have a magnetic moment of 1.73 BM, (spin only value)?

(A) O_2^+

(B) CuI

(C) $[Cu(NH_3)_4]Cl_2$

(D) O_2^-

Open solution

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Q9. MCQ (Single Correct Answer)

Which one of the following statements is not true about enzymes?

(A) Enzymes are non-specific for a reaction and substrate.

(B) Almost all enzymes are proteins.

(C) Enzymes work as catalysts by lowering the activation energy of a biochemical reaction.

(D) The action of enzymes is temperature and pH specific.

Open solution

Q10. MCQ (Single Correct Answer)

The hybridisations of the atomic orbitals of nitrogen in NO_2^-, NO_2^+ and NH_4^+ respectively are.

(A) sp^3, sp^2 and sp (B) sp, sp^2 and sp^3 (C) sp^3, sp and sp^2 (D) sp^2, sp and sp^3

Open solution

Q11. MCQ (Single Correct Answer)

Benzene on nitration gives nitrobenzene in presence of HNO_3 and H_2SO_4 mixture, where :

(A) both H_2SO_4 and HNO_3 act as a bases

(B) HNO_3 acts as an acid and H_2SO_4 acts as a base

(C) both H_2SO_4 and HNO_3 act as an acids

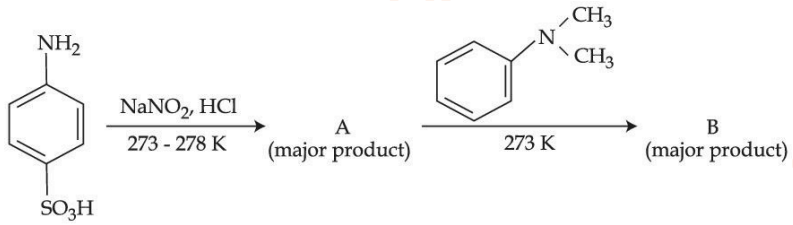
(D) HNO_3 acts as a base and H_2SO_4 acts as an acid

Open solution

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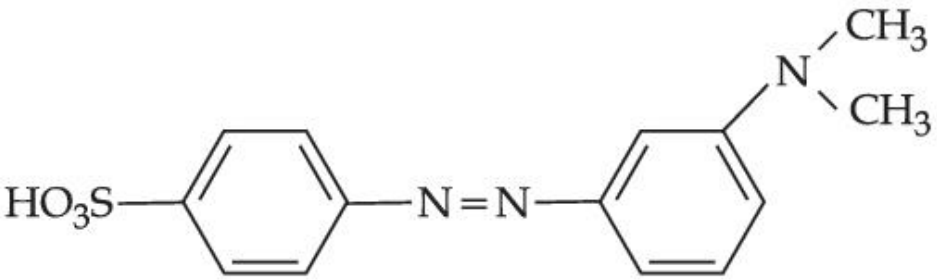
Q12. MCQ (Single Correct Answer)

Open solution

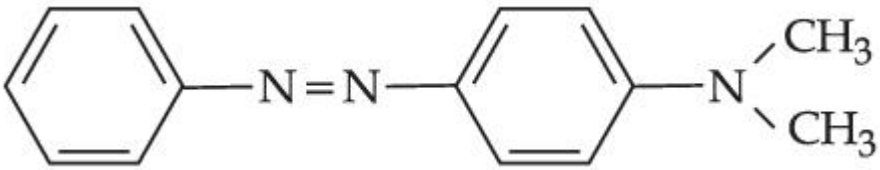


Consider the above reaction, compound B is :

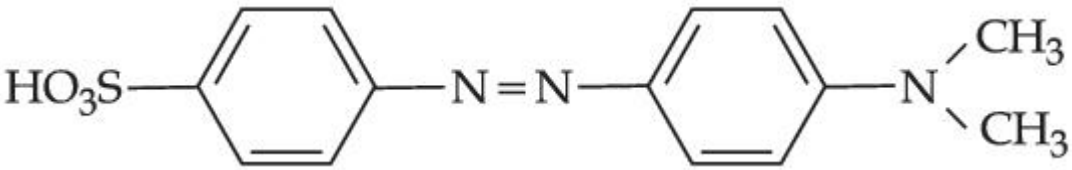
(A)



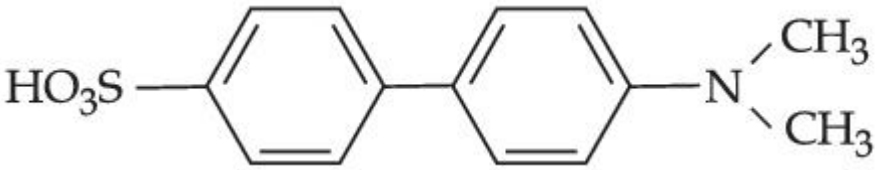
(B)



(C)

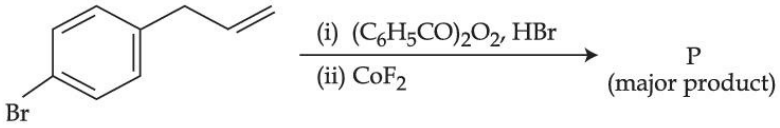


(D)



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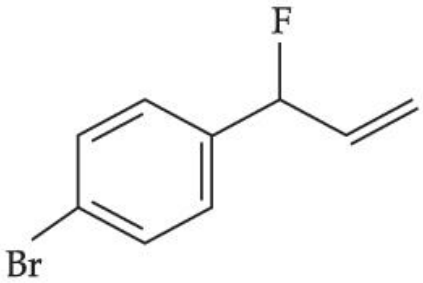
Q13. MCQ (Single Correct Answer)



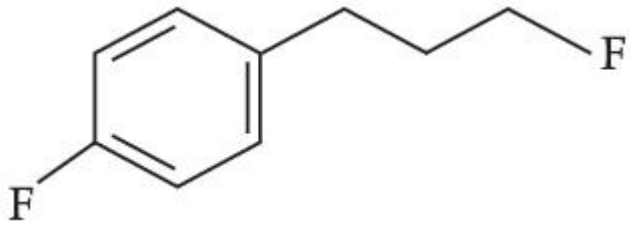
Open solution

Major product P of above reaction, is :

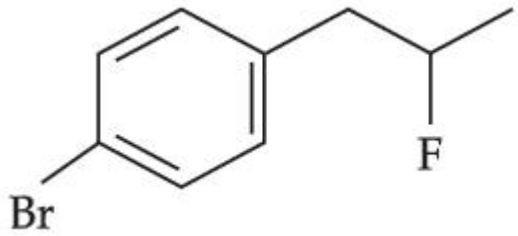
(A)



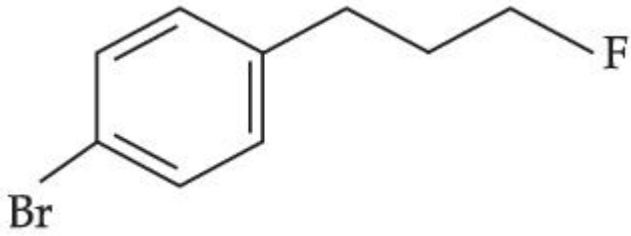
(B)



(C)



(D)



Q14. MCQ (Single Correct Answer)

Cu^{2+} salt reacts with potassium iodide to give

(A) Cu_2I_2 (B) Cu_2I_3 (C) CuI (D) $Cu(I_3)_2$

Open solution

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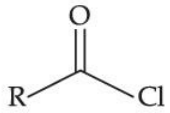
Q15. MCQ (Single Correct Answer)

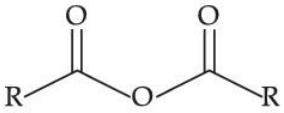
In Carius method, halogen containing organic compound is heated with fuming nitric acid in the presence of :

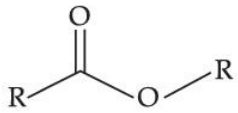
(A) HNO_3 **(B)** $AgNO_3$ **(C)** $CuSO_4$ **(D)** $BaSO_4$

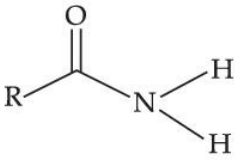
Open solution

Q16. MCQ (Single Correct Answer)

(A) 

(B) 

(C) 

(D) 

The correct order of their reactivity towards hydrolysis at room temperature is :

(A) (A) > (B) > (C) > (D) **(B)** (D) > (A) > (B) > (C)

(C) (D) > (B) > (A) > (C) **(D)** (A) > (C) > (B) > (D)

Open solution

Q17. Numerical

For a given chemical reaction A $\rightarrow$ B at 300 K the free energy change is -49.4 kJ mol^{-1} and the enthalpy of reaction is 51.4 kJ mol^{-1}. The entropy change of the reaction is _____ JK^{-1} mol^{-1}.

Open solution

Q18. Numerical

The wavelength of electrons accelerated from rest through a potential difference of 40 kV is $x \times 10^{-12}$ m. The value of x is _____. (Nearest integer)

Given : Mass of electron = 9.1×10^{-31} kg

Charge on an electron = 1.6×10^{-19} C

Planck's constant = 6.63×10^{-34} Js

Open solution

Q19. Numerical

The vapour pressures of A and B at 25°C are 90 mm Hg and 15 mm Hg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.6, then the mole fraction of B in the vapour phase is $x \times 10^{-1}$. The value of x is _____. (Nearest integer)

Open solution

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Q20. Numerical

4g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3. The value of x is _____ g. (Nearest integer)

Open solution

Q21. Numerical

When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _____. (Nearest integer)

[Atomic mass : Silver = 108, Bromine = 80]

Open solution

Q22. Numerical

$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$

In the above first order reaction the concentration of PCl_5 reduces from initial concentration 50 mol L^{-1} to 10 mol L^{-1} in 120 minutes at 300 K. The rate constant for the reaction at 300 K is $x \times 10^{-2}$ min^{-1}. The value of x is _____. [Given log5 = 0.6989]

Open solution

Q23. Numerical

An aqueous solution of $NiCl_2$ was heated with excess sodium cyanide in presence of strong oxidizing agent to form $[Ni(CN)_6]^{2-}$. The total change in number of unpaired electrons on metal centre is _____.

Open solution

Q24. Numerical

Potassium chlorate is prepared by electrolysis of KCl in basic solution as shown by following equation.

$6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$

A current of xA has to be passed for 10h to produce 10.0g of potassium chlorate. The value of x is _____. (Nearest integer)

(Molar mass of $KClO_3$ = 122.6 g mol^{-1}, F = 96500 C)

Open solution

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Mathematics 21 questions

Q25. MCQ (Single Correct Answer)

The value of $\tan\left(2\tan^{-1}\left(\frac{3}{5}\right)+\sin^{-1}\left(\frac{5}{13}\right)\right)$ is equal to :

(A) $\frac{-181}{69}$ (B) $\frac{220}{21}$ (C) $\frac{-291}{76}$ (D) $\frac{151}{63}$

Open solution

Q26. MCQ (Single Correct Answer)

The lines x = ay − 1 = z − 2 and x = 3y − 2 = bz − 2, (ab ≠ 0) are coplanar, if :

(A) b = 1, a∈R − {0} (B) a = 1, b∈R − {0}

(C) a = 2, b = 2 (D) a = 2, b = 3

Open solution

Q27. MCQ (Single Correct Answer)

If [x] denotes the greatest integer less than or equal to x, then the value of the integral $\int_{-\pi/2}^{\pi/2}[[x]-\sin x]dx$ is equal to :

(A) $-\pi$ (B) π (C) 0 (D) 1

Open solution

Q28. MCQ (Single Correct Answer)

If the real part of the complex number $(1-\cos\theta+2i\sin\theta)^{-1}$ is $\frac{1}{5}$ for $\theta\in(0,\pi)$, then the value of the integral $\int_0^{\theta}\sin x dx$ is equal to:

(A) 1 (B) 2 (C) −1 (D) 0

Open solution

Q29. MCQ (Single Correct Answer)

Let $f: R-\left\{\frac{\alpha}{6}\right\}\to R$ be defined by $f(x)=\frac{5x+3}{6x-\alpha}$. Then the value of α for which (fof)(x) = x, for all $x\in R-\left\{\frac{\alpha}{6}\right\}$, is :

(A) No such α exists (B) 5

(C) 8 (D) 6

Open solution

Q30. MCQ (Single Correct Answer)

If $f: R\to R$ is given by $f(x)=x+1$, then the value of $\lim_{n\to\infty}\frac{1}{n}\left[f(0)+f\left(\frac{5}{n}\right)+f\left(\frac{10}{n}\right)+\ldots\ldots+f\left(\frac{5(n-1)}{n}\right)\right]$ is :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{1}{2}$ (D) $\frac{7}{2}$

Open solution

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Q31. MCQ (Single Correct Answer)

Let A, B and C be three events such that the probability that exactly one of A and B occurs is (1 − k), the probability that exactly one of B and C occurs is (1 − 2k), the probability that exactly one of C and A occurs is (1 − k) and the probability of all A, B and C occur simultaneously is k^2, where 0 < k < 1. Then the probability that at least one of A, B and C occur is :

(A) greater than $\frac{1}{8}$ but less than $\frac{1}{4}$
(B) greater than $\frac{1}{2}$
(C) greater than $\frac{1}{4}$ but less than $\frac{1}{2}$
(D) exactly equal to $\frac{1}{2}$

Open solution

Q32. MCQ (Single Correct Answer)

The sum of all the local minimum values of the twice differentiable function f : R → R defined by $f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1)$ is :

(A) −22
(B) 5
(C) −27
(D) 0

Open solution

Q33. MCQ (Single Correct Answer)

Let y = y(x) satisfies the equation $\frac{dy}{dx} - |A| = 0$, for all x > 0, where $A = \begin{bmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{bmatrix}$. If $y(\pi) = \pi + 2$, then the value of $y\left(\frac{\pi}{2}\right)$ is :

(A) $\frac{\pi}{2} + \frac{4}{\pi}$
(B) $\frac{\pi}{2} - \frac{1}{\pi}$
(C) $\frac{3\pi}{2} - \frac{1}{\pi}$
(D) $\frac{\pi}{2} - \frac{4}{\pi}$

Open solution

Q34. MCQ (Single Correct Answer)

If the mean and variance of six observations 7, 10, 11, 15, a, b are 10 and $\frac{20}{3}$, respectively, then the value of | a − b | is equal to :

(A) 9
(B) 11
(C) 7
(D) 1

Open solution

Q35. MCQ (Single Correct Answer)

Let $g(t) = \int_{-\pi/2}^{\pi/2} \cos\left(\frac{\pi}{4}t + f(x)\right)dx$, where $f(x) = \log_e\left(x + \sqrt{x^2+1}\right), x \in R$. Then which one of the following is correct?

(A) g(1) = g(0)
(B) $\sqrt{2}g(1) = g(0)$
(C) $g(1) = \sqrt{2}g(0)$
(D) g(1) + g(0) = 0

Open solution

Q36. MCQ (Single Correct Answer)

Let P be a variable point on the parabola $y = 4x^2 + 1$. Then, the locus of the mid-point of the point P and the foot of the perpendicular drawn from the point P to the line y = x is :

(A) $(3x - y)^2 + (x - 3y) + 2 = 0$
(B) $2(3x - y)^2 + (x - 3y) + 2 = 0$
(C) $(3x - y)^2 + 2(x - 3y) + 2 = 0$
(D) $2(x - 3y)^2 + (3x - y) + 2 = 0$

Open solution

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Q37. MCQ (Single Correct Answer)

The value of k $\in$R, for which the following system of linear equations

$3x - y + 4z = 3$,

$x + 2y - 3z = -2$

$6x + 5y + kz = -3$,

has infinitely many solutions, is :

(A) 3 (B) -5 (C) 5 (D) -3

Open solution

Q38. MCQ (Single Correct Answer)

In a triangle ABC, if $\left|\overrightarrow{BC}\right| = 3$, $\left|\overrightarrow{CA}\right| = 5$ and $\left|\overrightarrow{BA}\right| = 7$, then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to :

(A) $\frac{19}{2}$ (B) $\frac{13}{2}$ (C) $\frac{11}{2}$ (D) $\frac{15}{2}$

Open solution

Q39. Numerical

The number of solutions of the equation

$\log_{(x+1)}(2x^2 + 7x + 5) + \log_{(2x+5)}(x+1)^2 - 4 = 0$, x > 0, is :

Open solution

Q40. Numerical

Let a curve y = y(x) be given by the solution of the differential equation $\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right)dx = \sqrt{e^{2x}-1}dy$. If it intersects y-axis at y = -1, and the intersection point of the curve with x-axis is $(\alpha, 0)$, then e^{α} is equal to _____.

Open solution

Q41. Numerical

For p > 0, a vector $\overrightarrow{v_2} = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\overrightarrow{v_1} = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about origin in counter clockwise direction. If $\tan\theta = \frac{\left(\alpha\sqrt{3}-2\right)}{\left(4\sqrt{3}+3\right)}$, then the value of α is equal to _____.

Open solution

Q42. Numerical

Consider a triangle having vertices A(-2, 3), B(1, 9) and C(3, 8). If a line L passing through the circum-centre of triangle ABC, bisects line BC, and intersects y-axis at point $\left(0, \frac{\alpha}{2}\right)$, then the value of real number α is _____.

Open solution

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Q43. Numerical

If the point on the curve $y^2 = 6x$, nearest to the point $\left(3, \frac{3}{2}\right)$ is (α, β), then $2(\alpha + \beta)$ is equal to _____.

Open solution

Q44. Numerical

Let a function g : [0, 4] $\rightarrow$ R be defined as

$g(x) = \begin{cases} \max\limits_{0 \le t \le x} \{t^3 - 6t^2 + 9t - 3\}, & 0 \le x \le 3 \\ 4 - x, & 3 < x \le 4 \end{cases}$, then the number of points in the interval (0, 4) where g(x) is NOT differentiable, is _____.

Open solution

Q45. Numerical

If $\lim\limits_{x \to 0} \frac{\alpha x e^x - \beta \log_e(1+x) + \gamma x^2 e^{-x}}{x \sin^2 x} = 10, \alpha, \beta, \gamma \in R$, then the value of $\alpha + \beta + \gamma$ is _____.

Open solution

Physics 27 questions

Q46. MCQ (Single Correct Answer)

If the Kinetic energy of a moving body becomes four times its initial Kinetic energy, then the percentage change in its momentum will be :

(A) 100% (B) 200% (C) 300% (D) 400%

Open solution

Q47. MCQ (Single Correct Answer)

A boy reaches the airport and finds that the escalator is not working. He walks up the stationary escalator in time t_1. If he remains stationary on a moving escalator then the escalator takes him up in time t_2. The time taken by him to walk up on the moving escalator will be :

(A) $\frac{t_1 t_2}{t_2 - t_1}$ (B) $\frac{t_1 + t_2}{2}$

(C) $\frac{t_1 t_2}{t_2 + t_1}$ (D) $t_2 - t_1$

Open solution

Q48. MCQ (Single Correct Answer)

A satellite is launched into a circular orbit of radius R around earth, while a second satellite is launched into a circular orbit of radius 1.02 R. The percentage difference in the time periods of the two satellites is :

(A) 1.5 (B) 2.0 (C) 0.7 (D) 3.0

Open solution

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Q49. MCQ (Single Correct Answer)

With what speed should a galaxy move outward with respect to earth so that the sodium-D line at wavelength 5890 $\overset{o}{A}$ is observed at 5896 $\overset{o}{A}$?

(A) 306 km/sec (B) 322 km/sec (C) 296 km/sec (D) 336 km/sec

Open solution

Q50. MCQ (Single Correct Answer)

The length of a metal wire is l_1, when the tension in it is T_1 and is l_2 when the tension is T_2. The natural length of the wire is :

(A) $\sqrt{l_1 l_2}$

(B) $\frac{l_1T_2-l_2T_1}{T_2-T_1}$

(C) $\frac{l_1T_2+l_2T_1}{T_2+T_1}$

(D) $\frac{l_1+l_2}{2}$

Open solution

Q51. MCQ (Single Correct Answer)

In an electromagnetic wave the electric field vector and magnetic field vector are given as $\overrightarrow{E} = E_0\hat{i}$ and $\overrightarrow{B} = B_0\hat{k}$ respectively. The direction of propagation of electromagnetic wave is along :

(A) $\left(\hat{k}\right)$ (B) $\hat{j}$ (C) $\left(-\hat{k}\right)$ (D) $\left(-\hat{j}\right)$

Open solution

Q52. MCQ (Single Correct Answer)

For a series LCR circuit with R = 100 Ω, L = 0.5 mH and C = 0.1 pF connected across 220V$-$50 Hz AC supply, the phase angle between current and supplied voltage and the nature of the circuit is :

(A) 0°, resistive circuit

(B) $\approx$ 90°, predominantly inductive circuit

(C) 0°, resonance circuit

(D) $\approx$ 90°, predominantly capacitive circuit

Open solution

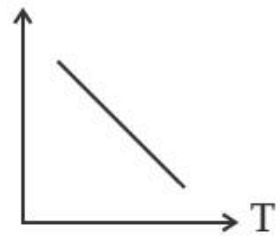
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Q53. MCQ (Single Correct Answer)

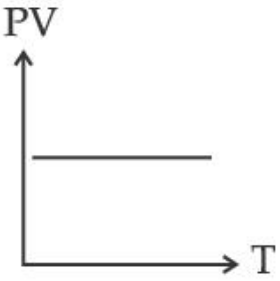
Open solution

Which of the following graphs represent the behavior of an ideal gas? Symbols have their usual meaning.

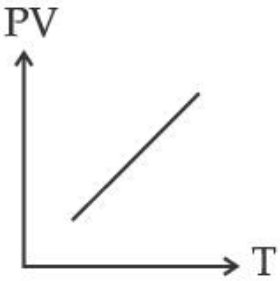
(A)



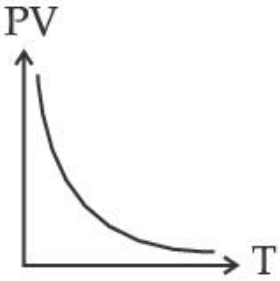
(B)



(C)



(D)



Q54. MCQ (Single Correct Answer)

Open solution

A particle is making simple harmonic motion along the X-axis. If at a distances x_1 and x_2 from the mean position the velocities of the particle are v_1 and v_2 respectively. The time period of its oscillation is given as :

(A) $T = 2\pi\sqrt{\frac{x_2^2+x_1^2}{v_1^2-v_2^2}}$

(B) $T = 2\pi\sqrt{\frac{x_2^2+x_1^2}{v_1^2+v_2^2}}$

(C) $T = 2\pi\sqrt{\frac{x_2^2-x_1^2}{v_1^2+v_2^2}}$

(D) $T = 2\pi\sqrt{\frac{x_2^2-x_1^2}{v_1^2-v_2^2}}$

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Q55. MCQ (Single Correct Answer)

An electron having de-Broglie wavelength λ is incident on a target in a X-ray tube. Cut-off wavelength of emitted X-ray is :

(A) 0

(B) $\frac{2m^2c^2\lambda^2}{h^2}$

(C) $\frac{2mc\lambda^2}{h}$

(D) $\frac{hc}{mc}$

Open solution

Q56. MCQ (Single Correct Answer)

A body rolls down an inclined plane without slipping. The kinetic energy of rotation is 50% of its translational kinetic energy. The body is :

(A) Solid sphere (B) Solid cylinder (C) Hollow cylinder (D) Ring

Open solution

Q57. MCQ (Single Correct Answer)

If time (t), velocity (v), and angular momentum (l) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v and l is :

(A) $[t^{-1}v^1l^{-2}]$ (B) $[t^1v^2l^{-1}]$ (C) $[t^{-2}v^{-1}l^1]$ (D) $[t^{-1}v^{-2}l^1]$

Open solution

Q58. MCQ (Single Correct Answer)

The correct relation between the degrees of freedom f and the ratio of specific heat γ is :

(A) $f = \frac{2}{\gamma-1}$ (B) $f = \frac{2}{\gamma+1}$ (C) $f = \frac{\gamma+1}{2}$ (D) $f = \frac{1}{\gamma+1}$

Open solution

Q59. MCQ (Single Correct Answer)

Consider a binary star system of star A and star B with masses m_A and m_B revolving in a circular orbit of radii r_A an r_B, respectively. If T_A and T_B are the time period of star A and star B, respectively,

Then :

(A) $\frac{T_A}{T_B} = \left(\frac{r_A}{r_B}\right)^{\frac{3}{2}}$

(B) $T_A = T_B$

(C) $T_A > T_B$ (if $m_A > m_B$)

(D) $T_A > T_B$ (if $r_A > r_B$)

Open solution

Q60. MCQ (Single Correct Answer)

A body at rest is moved along a horizontal straight line by a machine delivering a constant power. The distance moved by the body in time 't' is proportional to :

(A) $t^{\frac{3}{2}}$ (B) $t^{\frac{1}{2}}$ (C) $t^{\frac{1}{4}}$ (D) $t^{\frac{3}{4}}$

Open solution

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Q61. MCQ (Single Correct Answer)

Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P}+\vec{Q}$ is n times the magnitude of $\vec{P}-\vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is :

(A) $\sin^{-1}\left(\frac{n-1}{n+1}\right)$ (B) $\cos^{-1}\left(\frac{n-1}{n+1}\right)$

(C) $\sin^{-1}\left(\frac{n^2-1}{n^2+1}\right)$ (D) $\cos^{-1}\left(\frac{n^2-1}{n^2+1}\right)$

Open solution

Q62. MCQ (Single Correct Answer)

Two small drops of mercury each of radius R coalesce to form a single large drop. The ratio of total surface energy before and after the change is :

(A) $2^{\frac{1}{3}}:1$ (B) $1:2^{\frac{1}{3}}$ (C) 2 : 1 (D) 1 : 2

Open solution

Q63. MCQ (Single Correct Answer)

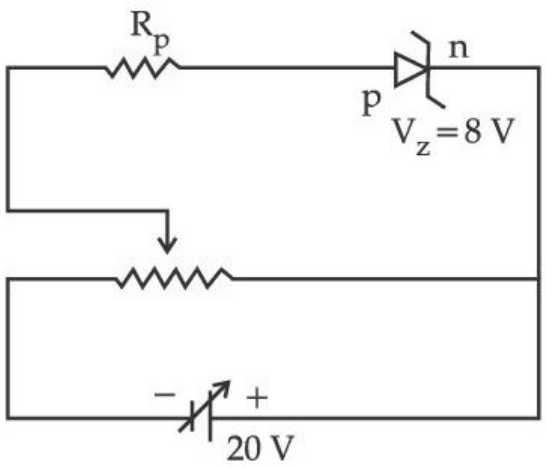
The magnetic susceptibility of a material of a rod is 499. Permeability in vacuum is $4\pi \times 10^{-7}$ H/m. Absolute permeability of the material of the rod is :

(A) $4\pi \times 10^{-4}$ H/m (B) $2\pi \times 10^{-4}$ H/m (C) $3\pi \times 10^{-4}$ H/m (D) $\pi \times 10^{-4}$ H/m

Open solution

Q64. Numerical

A zener diode having zener voltage 8 V and power dissipation rating of 0.5 W is connected across a potential divider arranged with maximum potential drop across zener diode is as shown in the diagram. The value of protective resistance R_p is ..... Ω.



Open solution

Q65. Numerical

A body of mass 'm' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent. The value of x is _____.

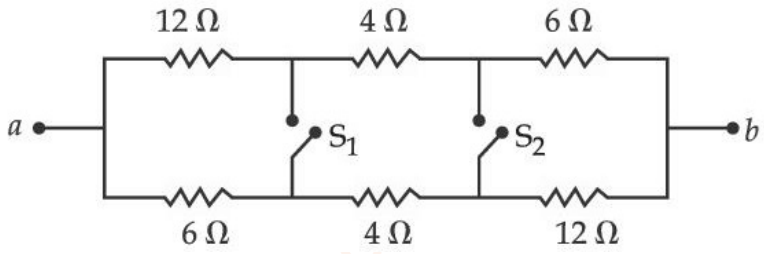
Open solution

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Q66. Numerical

In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____ Ω.

Open solution



Q67. Numerical

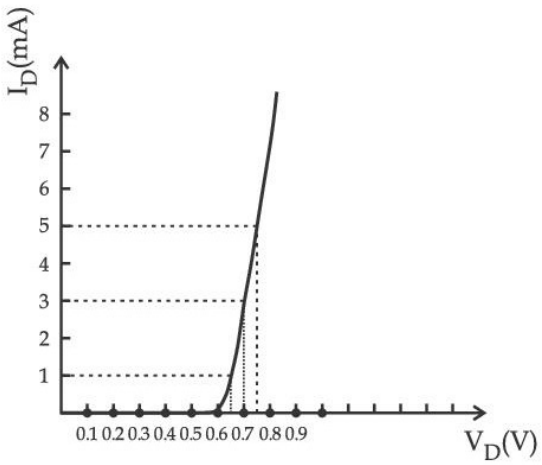
Two bodies, a ring and a solid cylinder of same material are rolling down without slipping an inclined plane. The radii of the bodies are same. The ratio of velocity of the centre of mass at the bottom of the inclined plane of the ring to that of the cylinder is $\frac{\sqrt{x}}{2}$. Then, the value of x is _____.

Open solution

Q68. Numerical

For the forward biased diode characteristics shown in the figure, the dynamic resistance at I_D = 3 mA will be _____ Ω.

Open solution



Q69. Numerical

A series LCR circuit of R = 5Ω, L = 20 mH and C = 0.5 μF is connected across an AC supply of 250 V, having variable frequency. The power dissipated at resonance condition is _____ $\times 10^2$ W.

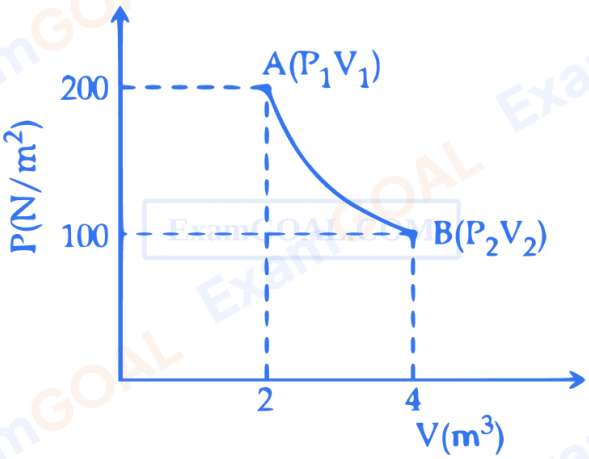
Open solution

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Q70. Numerical

One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be _____ $\times 10^{-1}$ J. [Given : R = 8.3 J/mole K, ln2 = 0.6931] (Round off to the nearest integer)

Open solution



Q71. Numerical

A certain metallic surface is illuminated by monochromatic radiation of wavelength λ. The stopping potential for photoelectric current for this radiation is $3V_0$. If the same surface is illuminated with a radiation of wavelength 2λ, the stopping potential is V_0. The threshold wavelength of this surface for photoelectric effect is _____ λ.

Open solution

Q72. Numerical

A body rotating with an angular speed of 600 rpm is uniformly accelerated to 1800 rpm in 10 sec. The number of rotations made in the process is _____.

Open solution

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q73. MCQ (Single Correct Answer) · **Chemistry**

The single largest industrial application of dihydrogen is :

(A) Manufacture of metal hybrides

(B) Rocket fuel in space research

(C) In the synthesis of ammonia

(D) In the synthesis of nitric acid

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Consider two chemical reactions (A) and (B) that take place during metallurgical process:

(A) $ZnCO_{3(s)} \xrightarrow{\Delta} ZnO_{(s)} + CO_{2(g)}$

(B) $2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}$

The correct option of names given to them respectively is :

(A) (A) is calcination and (B) is roasting

(B) Both (A) and (B) are producing same product so both are roasting

(C) Both (A) and (B) are producing same product so both are calcination

(D) (A) is roasting and (B) is calcination

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Bakelite is a cross-linked polymer of formaldehyde and :

(A) PHBV **(B)** Buna-S **(C)** Novolac **(D)** Dacron

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following gases is reported to retard photosynthesis?

(A) CO **(B)** CFCs **(C)** CO_2 **(D)** NO_2

Open solution

Q77. Numerical · **Chemistry**

100 ml of 0.0018% (w/v) solution of Cl^- ion was the minimum concentration of Cl^- required to precipitate a negative sol in one h. The coagulating value of Cl^- ion is _____ (Nearest integer)

Open solution

Q78. Numerical · **Chemistry**

Diamond has a three dimensional structure of C atoms formed by covalent bonds. The structure of diamond has face centred cubic lattice where 50% of the tetrahedral voids are also occupied by carbon atoms. The number of carbon atoms present per unit cell of diamond is _____.

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

For the natural numbers m, n, if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \ldots + a_{m+n}y^{m+n}$ and $a_1 = a_2 = 10$, then the value of (m + n) is equal to :

(A) 88 **(B)** 64 **(C)** 100 **(D)** 80

Open solution

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Q80. MCQ (Single Correct Answer) · **Mathematics**

Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$. If $\frac{r_1}{r_2} = a + b\sqrt{2}$, then a + b is equal to :

Open solution

(A) 3 (B) 11 (C) 5 (D) 7

Q81. MCQ (Single Correct Answer) · **Mathematics**

Consider the following three statements :

(A) If 3 + 3 = 7 then 4 + 3 = 8

(B) If 5 + 3 = 8 then earth is flat.

(C) If both (A) and (B) are true then 5 + 6 = 17.

Then, which of the following statements is correct?

Open solution

(A) (A) is false, but (B) and (C) re true (B) (A) and (C) are true while (B) is false

(C) (A) is true while (B) and (C) are false (D) (A) and (B) are false while (C) is true

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let in a right angled triangle, the smallest angle be θ. If a triangle formed by taking the reciprocal of its sides is also a right angled triangle, then $\sin\theta$ is equal to :

Open solution

(A) $\frac{\sqrt{5}+1}{4}$ (B) $\frac{\sqrt{5}-1}{2}$ (C) $\frac{\sqrt{2}-1}{2}$ (D) $\frac{\sqrt{5}-1}{4}$

Q83. MCQ (Single Correct Answer) · **Mathematics**

Consider the line L given by the equation

$\frac{x-3}{2} = \frac{y-1}{1} = \frac{z-2}{1}$.

Open solution

Let Q be the mirror image of the point $(2, 3, -1)$ with respect to L. Let a plane P be such that it passes through Q, and the line L is perpendicular to P. Then which of the following points is on the plane P?

(A) $(-1, 1, 2)$ (B) (1, 1, 1) (C) (1, 1, 2) (D) (1, 2, 2)

Q84. MCQ (Single Correct Answer) · **Mathematics**

If sum of the first 21 terms of the series $\log_{9^{1/2}}x + \log_{9^{1/3}}x + \log_{9^{1/4}}x + \ldots\ldots$, where x > 0 is 504, then x is equal to

Open solution

(A) 243 (B) 9 (C) 7 (D) 81

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Q85. Numerical · **Mathematics**

Let $A = \{a_{ij}\}$ be a 3 × 3 matrix,

where $a_{ij} = \begin{cases} (-1)^{j-i} & if \quad i < j, \\ 2 & if \quad i = j, \\ (-1)^{i+j} & if \quad i > j \end{cases}$

then $\det(3Adj(2A^{-1}))$ is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

For k ∈ N, let $\frac{1}{\alpha(\alpha+1)(\alpha+2).....(\alpha+20)} = \sum_{k=0}^{20} \frac{A_k}{\alpha+k}$, where $\alpha > 0$. Then the value of $100\left(\frac{A_{14}+A_{15}}{A_{13}}\right)^2$ is equal to _____.

Open solution

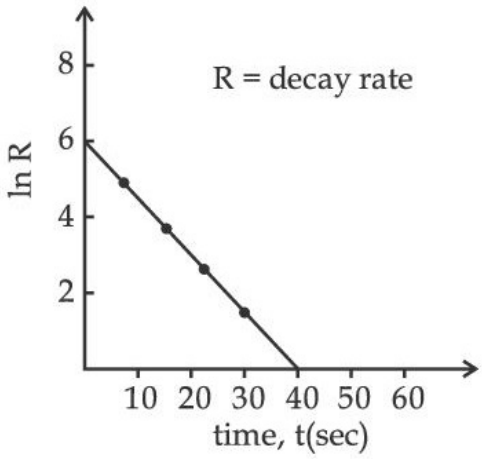
Q87. Numerical · **Mathematics**

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of $47 \sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ is equal to _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

For a certain radioactive process the graph between ln R and t(sec) is obtained as shown in the figure. Then the value of half life for the unknown radioactive material is approximately :



(A) 9.15 sec (B) 6.93 sec (C) 2.62 sec (D) 4.62 sec

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

At an angle of 30° to the magnetic meridian, the apparent dip is 45°. Find the true dip :

(A) $\tan^{-1}\sqrt{3}$ (B) $\tan^{-1}\frac{1}{\sqrt{3}}$

(C) $\tan^{-1}\frac{2}{\sqrt{3}}$ (D) $\tan^{-1}\frac{\sqrt{3}}{2}$

Open solution

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Q90. Numerical · **Physics**

A radioactive substance decays to $\left(\frac{1}{16}\right)^{th}$ of its initial activity in 80 days. The half life of the radioactive substance expressed in days is _____.

Open solution

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Answer Key

Chemistry

1. **D**	2. **D**	3. **B**	4. **D**	5. **D**
6. **B**	7. **C**	8. **B**	9. **A**	10. **D**
11. **D**	12. **C**	13. **D**	14. **C**	15. **B**
16. **A**	17. **336**	18. **6**	19. **1**	20. **1**
21. **68**	22. **1**	23. **2**	24. **1**	

Mathematics

25. **B**	26. **A**	27. **A**	28. **A**	29. **B**
30. **D**	31. **B**	32. **C**	33. **A**	34. **D**
35. **B**	36. **B**	37. **B**	38. **C**	39. **1**
40. **2**	41. **6**	42. **9**	43. **9**	44. **1**
45. **3**				

Physics

46. **A**	47. **C**	48. **D**	49. **A**	50. **B**
51. **D**	52. **D**	53. **C**	54. **D**	55. **C**
56. **B**	57. **D**	58. **A**	59. **B**	60. **A**
61. **D**	62. **A**	63. **B**	64. **192**	65. **3**
66. **10**	67. **3**	68. **25**	69. **125**	70. **17258**
71. **4**	72. **200**			

Out of syllabus

73. **C**	74. **A**	75. **C**	76. **D**	77. **1**
78. **8**	79. **D**	80. **C**	81. **B**	82. **B**
83. **D**	84. **D**	85. **108**	86. **9**	87. **7**
88. **D**	89. **D**	90. **20**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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